

1 Product name

UrbanDrop

An intelligent communal parking management system designed to provide designated legal parking locations for delivery personnel in high-impact urban areas.

2. Problem statement

Since the COVID-19 pandemic, delivery platforms such as DoorDash and Uber Eats have grown rapidly, increasing the number of vehicles operating in already congested urban areas where legal short-term parking is limited.

Under time pressure, drivers often stop in unauthorized zones, including bus lanes and restricted areas, contributing to congestion and exposing them to parking violations and financial penalties.

Current navigation applications prioritize route efficiency but do not coordinate delivery-specific legal parking availability. As a result, no dedicated system exists to manage lawful short-term parking for urban delivery operations.

3. Product objectives

1. To provide delivery personnel with clear and accessible information about designated legal parking locations.
2. To reduce traffic congestion caused by improper delivery vehicle parking in urban areas.
3. To increase awareness and utilization of designated legal short-term parking spaces.

4. Functional requirements

FR-1: User Authentication

The system shall allow registered delivery personnel and authorized administrators to have role-based access to the application.

FR-2: Parking Spot Management

The system shall allow authorized administrators to create, update, or deactivate communal parking spot records.

FR-3: Communal Parking Spot Display

The system shall display designated communal legal short-term parking spots on a map interface.

FR-4: Parking Spot Information

The system shall provide detailed information for each communal parking spot, including location, permitted parking duration, usage rules, and operational hours.

FR-5: High-Impact Area Identification

The system shall identify and visually highlight high-impact urban areas based on aggregated parking violation data.

FR-6: Navigation to Parking Spots

The system shall allow delivery personnel to obtain directions from their current location to a selected legal parking spot.

5. Nonfunctional requirements

NFR-1: Performance

The system shall retrieve and display all communal parking spots within a 1-kilometer radius of the user's delivery destination within 2 seconds for 95% of requests under normal operating conditions.

NFR-2: Update Frequency

The system shall refresh communal parking spot information and high-impact area indicators at least every 24 hours.

NFR-3: Data Accuracy

The system shall maintain consistency between displayed parking spot restrictions and the latest administrator-published records, and shall display the "last updated" timestamp for each parking spot.

NFR-4: Reliability

If real-time map services are unavailable, the system shall provide a fallback list view of nearby communal parking spots sorted by distance.

6. Use cases

